Computer Vision

DPS920

Session 11

Overview

Pandas
ML in general
Type of ML algorithms
Datasets
KNN
Distance metrics
Iris data classification

Agenda

Preprocessing pipeline
Diabetes classification

Steps to Solve ML Problems

Data and Preprocessing

ML Algorithm

Plot and Evaluate

Data Preprocessing

- Read data accurately
- Convert to numerical values
- Encoding categorical values
- Replace Null values
- Normalize values
- Separate features and labels
- Splitting data

```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score


dataset = pd.read_csv("S12_diabetes.csv")

zero_not_accepted = ["Glucose", "BloodPressure", "SkinThickness", "Insulin", "BMI"]
for column in zero_not_accepted:
    dataset[column] = dataset[column].replace(0, np.nan)
    mean = int(dataset[column].mean(skipna=True))
    dataset[column] = dataset[column].replace(np.nan, mean)

X = dataset.iloc[:, :8]
y = dataset.iloc[:, 8]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)


clf = KNeighborsClassifier(n_neighbors=11)
clf.fit(X_train, y_train)

y_pred = clf.predict(X_test)
acc = accuracy_score(y_test, y_pred)
print(f'accuracy: {acc}')
```

Normalization

Prevent features with larger ranges from dominating

Base comparison in distance based models

Standard scaler

Min-Max scaler

Standard Scaler

- Standard Scaler transforms features to have zero mean and unit variance using:

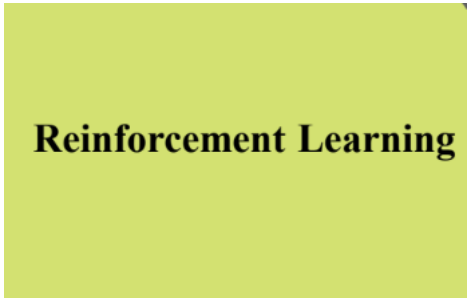
$$z = \frac{x - \mu}{\sigma}$$

$\mu =$ Mean
$\sigma =$ Standard Deviation

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

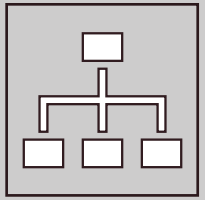
```python
from sklearn.preprocessing import StandardScaler

data = [[0, 0],
        [0, 0],
        [1, 1],
        [1, 1]]

scaler = StandardScaler()
new_data = scaler.fit_transform(data)
print(new_data)
```

ML Algorithm Categorizations



ML Algorithm Categorizations

Supervised Learning Algorithms Categorization



Classification

Regression

Regression

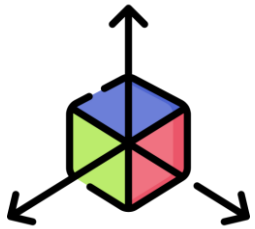
i) Simple

linear

nonlinear

i) Multiple

linear

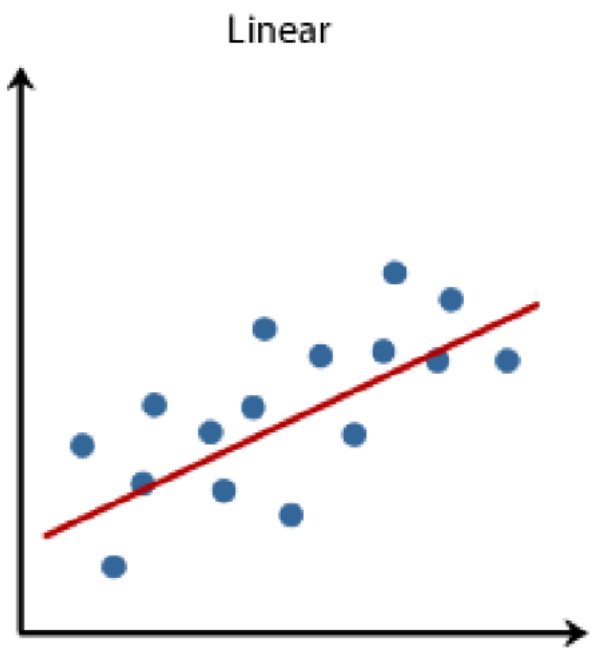


Polynomial

nonlinear

Simple Linear Regression

model the relationship between one independent variable (X) and one dependent variable (Y)

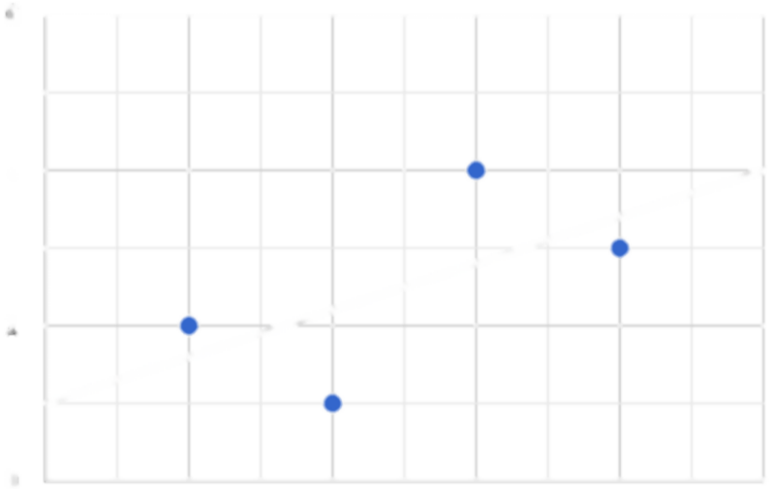


intercept

$$y = a + bx$$

slope

Problem



Our goal:

find a line that minimizes our error

Solution 1

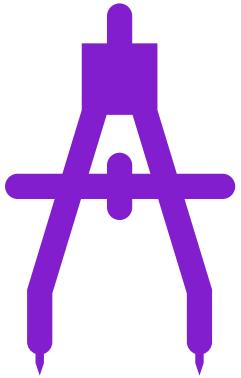
Visualize data and plot points → not scalable

Use Closed-Form Formulas → complex

Use optimization algorithms

Formulate

Points: (1,2), (2,1), (3,4), (4,3)



Solution: map to a linear line

Solution

- Points: (1,2), (2,1), (3,4), (4,3)

$$\left.\begin{aligned} a*1 + b &= 2 \\ a*2 + b &= 1 \\ a*3 + b &= 4 \\ a*4 + b &= 3 \end{aligned}\right\}$$

Can't find the exact **linear** line

Solution

- Points: (1,2), (2,1), (3,4), (4,3)

$$\left.\begin{aligned} a*1 + b &= 2 \\ a*2 + b &= 1 \\ a*3 + b &= 4 \\ a*4 + b &= 3 \end{aligned}\right\}$$

We want to find a line to minimize the distance between the real value and the lines output

$$\begin{array}{cccc} y1, & y2, & y3, & y4 \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ y1', & y2', & y3', & y4' \end{array}$$

Solution

- Points: (1,2), (2,1), (3,4), (4,3)

$$\left.\begin{aligned} a*1 + b &= 2 \\ a*2 + b &= 1 \\ a*3 + b &= 4 \\ a*4 + b &= 3 \end{aligned}\right\}$$

We want to find a line to minimize the distance between the real value and the lines output

$$\begin{array}{cccc} y1, & y2, & y3, & y4 \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ y1', & y2', & y3', & y4' \end{array} \longrightarrow \text{Minimize the distance}$$

Solution

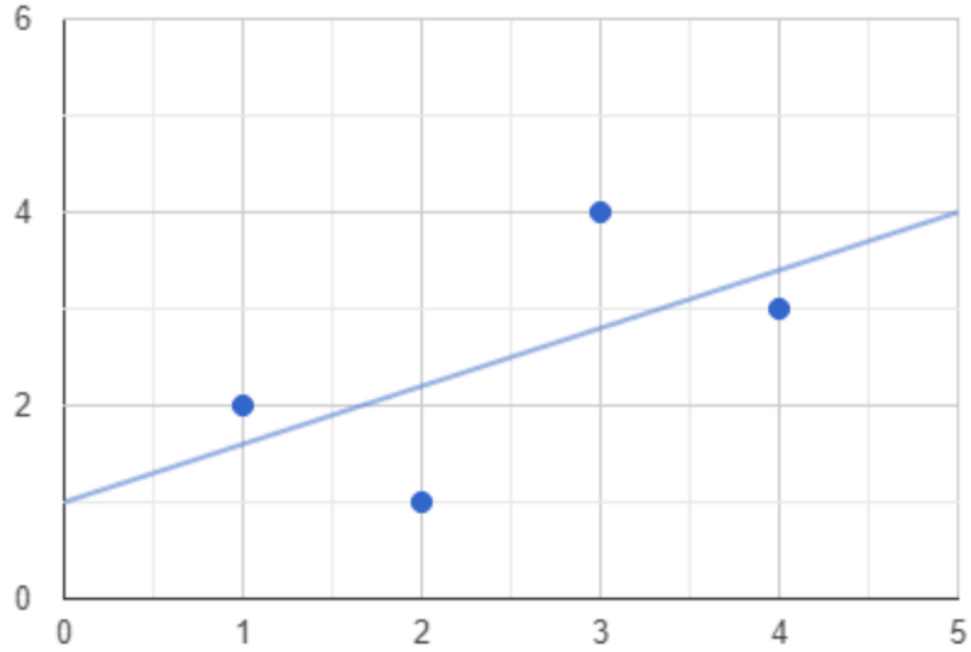
- Points: (1,2), (2,1), (3,4), (4,3)

$$\left.\begin{aligned} a*1 + b &= 2 \\ a*2 + b &= 1 \\ a*3 + b &= 4 \\ a*4 + b &= 3 \end{aligned}\right\}$$

We want to find a line to minimize the distance between the real value and the lines output

$$\begin{array}{cccc} y1, & y2, & y3, & y4 \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ y1', & y2', & y3', & y4' \end{array} \longrightarrow \min \left[(y1-y1')^2 + (y2-y2')^2 + (y3-y3')^2 + (y4-y4')^2 \right]$$

Example



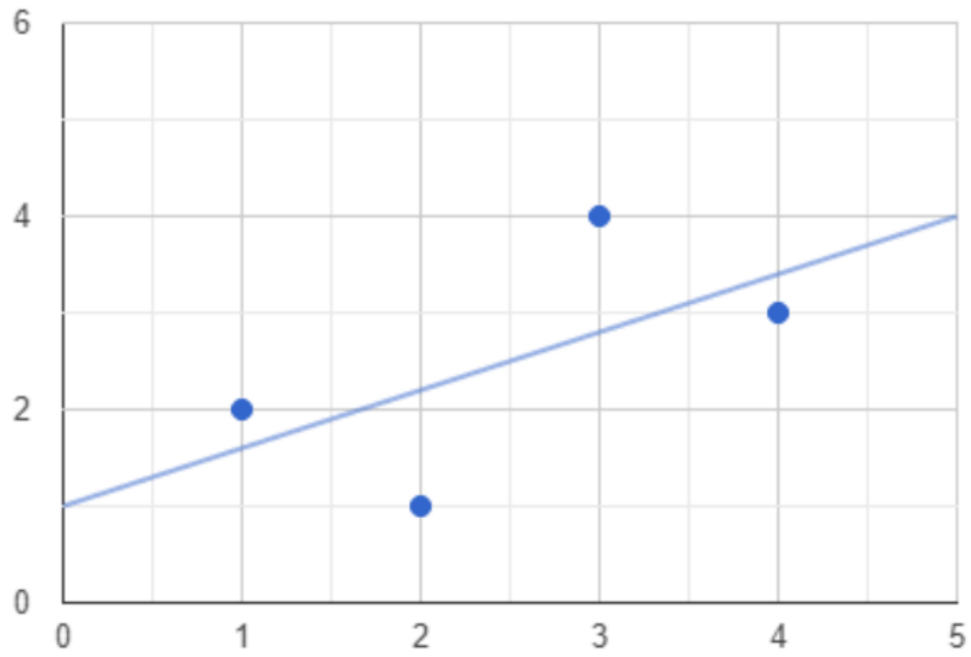
Actual outputs: 2, 1, 4, 3

Line's outputs: -1, 0, 1, 2

We do not know the equation of the line! But we want to find it

$y' = ax + b$

Example



Actual outputs: 2, 1, 4, 3

$$(y1-y1')^2 + (y2-y2')^2 + (y3-y3')^2 + (y4-y4')^2$$

$$\text{loss/distance} = (2-(a+b))^2 + (1-(2a+b))^2 + (4-(3a+b))^2 + (3-(4a+b))^2$$

Example

$$\text{loss/distance} = (2-(a+b))^2 + (1-(2a+b))^2 + (4-(3a+b))^2 + (3-(4a+b))^2$$

$$= 30*a^2 + 20*a*b + 56*a + 4*b^2 - 20*b + 30$$

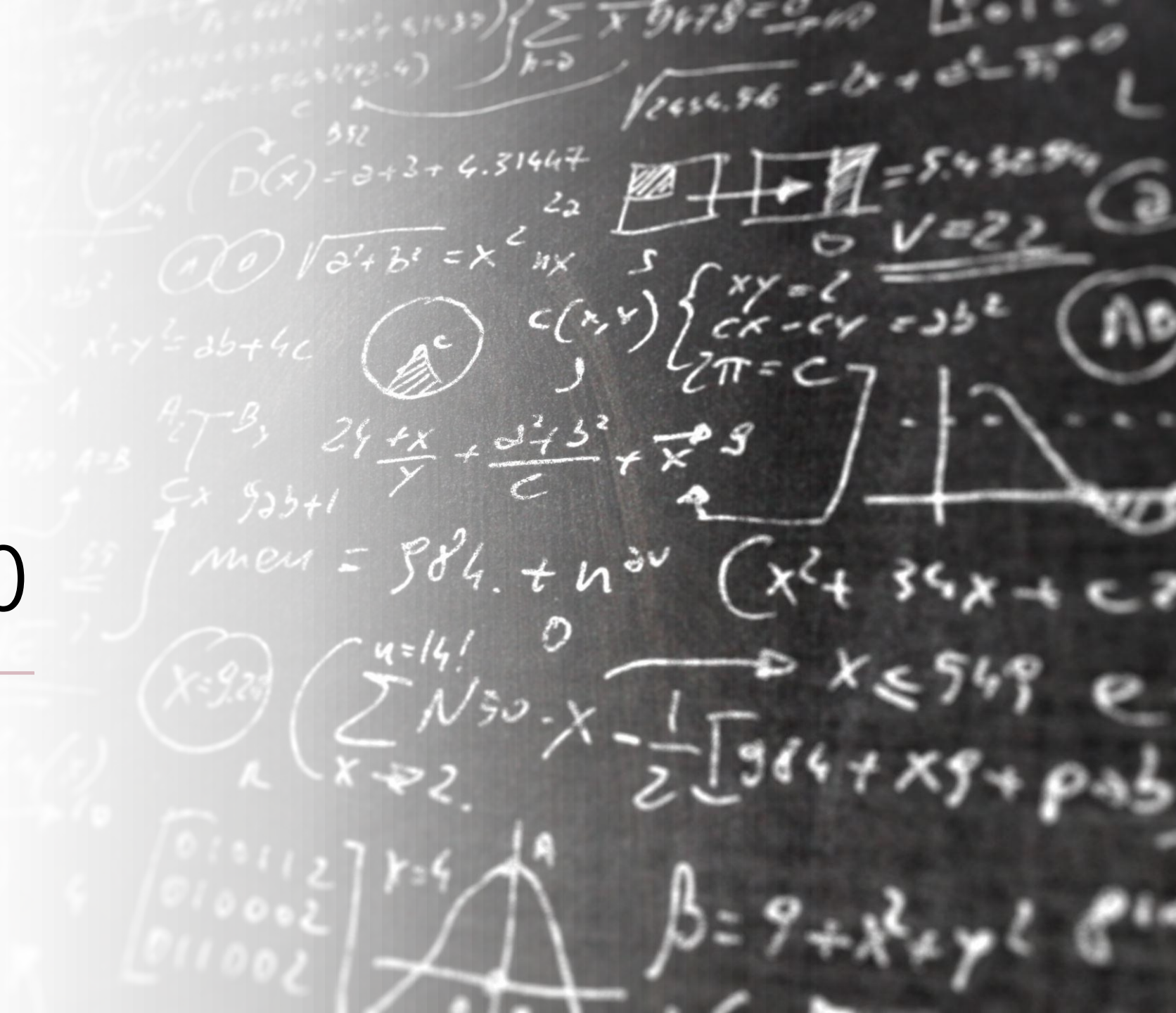
Minimum Loss

- We want to move towards a point with minimum loss!
- It is not the best fit, but it is better than nothing!

Minimum Loss

- Mathematically speaking, when we want the minimum of something what did we do?

Derivative = 0



Equation

- $30*a^2 + 20*a*b + 56*a + 4*b^2 - 20*b + 30$

$df/da = 0$

$df/db = 0$

Solve Equation

- $30*a^2 + 20*a*b + 56*a + 4*b^2 - 20*b + 30$

$df/da = 0$

$df/db = 0$

$60a + 20b + 56 = 0$

$20a + 8b + 20 = 0$

Solve Equation

- $30*a^2 + 20*a*b + 56*a + 4*b^2 - 20*b + 30$

$df/da = 0$

$df/db = 0$

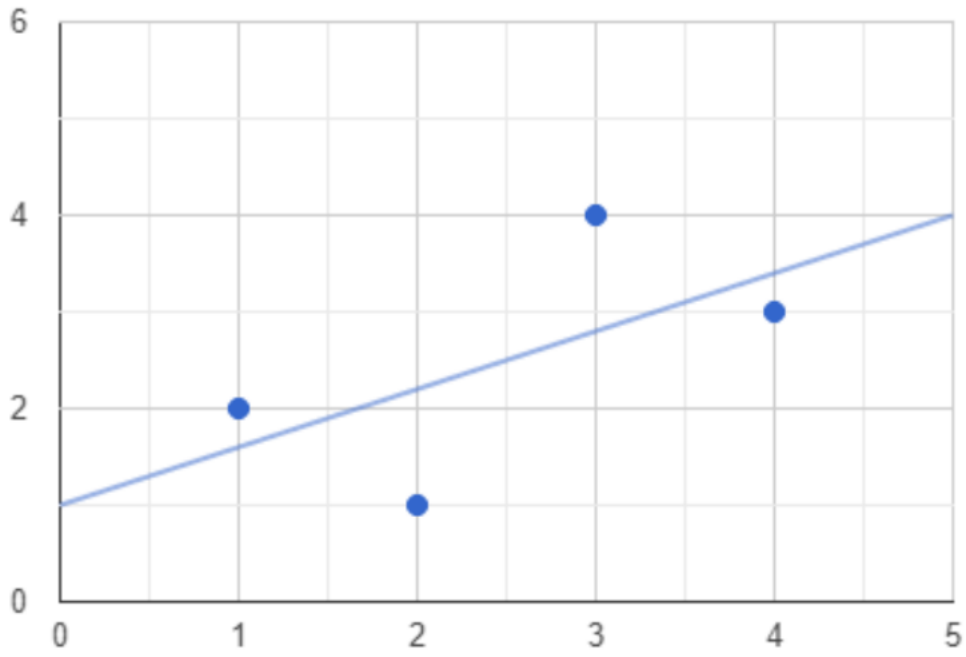
$$\boxed{\begin{array}{l} 60a + 20b + 56 = 0 \\ \\ 20a + 8b + 20 = 0 \end{array}} \Rightarrow \begin{array}{l} a = 1 \\ b = 0.6 \end{array}$$

Final Equation

Now we have a line with minimum error

$$y = x + 0.6$$

Final Output



$y = x + 0.6$

```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error


dataset = pd.read_csv("S12_student_scores.csv")

X = dataset.iloc[:, :-1]
y = dataset.iloc[:, 1]

plt.scatter(X, y)
plt.title("Hours vs Marks")
plt.xlabel("Hours")
plt.ylabel("Marks")
plt.show()

X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=42)

regressor = LinearRegression()
regressor.fit(X_train, y_train)

print(regressor.coef_)
print(regressor.intercept_)

y_pred = regressor.predict(X_test)
df = pd.DataFrame({'Actual':  y_test, "prediction": y_pred})

print(f"MAE: {mean_absolute_error(y_test, y_pred)}")
print(f"MAE: {mean_squared_error(y_test, y_pred)}")
```